

Trainings ▾

Preparatory Steps

Drives Alternator

⚠ CAUTION ⚠

The belt tensioner is spring loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner or personal injury.

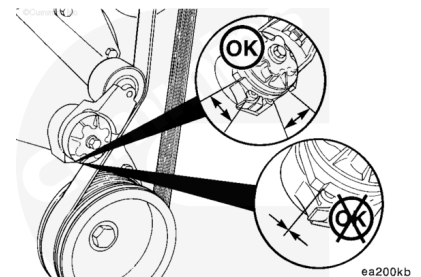
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Remove the cooling fan belt. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/10/10-008-062.html>).

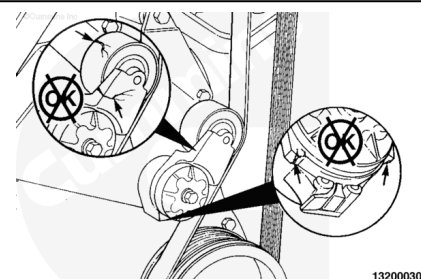
Initial Check

Without EGR

With the engine turned off and with the water pump/fan belt installed, check that neither the top nor bottom tensioner arm stop is touching the cast boss on the tensioner body. If either of the stops is touching a boss, the water pump/fan belt **must** be replaced. Check to make sure the correct belt part number is being used if either condition exists.



Check the tensioner pulley and body for cracks. If any cracks are noticed, the tensioner **must** be replaced.

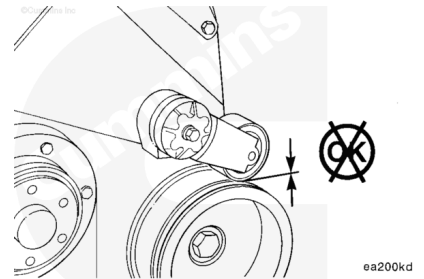


Using a 1/2-inch square drive, lift belt tensioner to relieve tension on the belt.

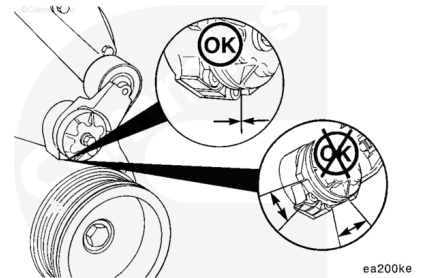
Remove the water pump/fan belt.

Note : The belt tensioner is spring-loaded and **must** be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner.

If the tensioner pulley touches the accessory drive pulley after the tensioner has been fully relaxed, the bottom tensioner arm stop boss has broken and the tensioner **must** be replaced.



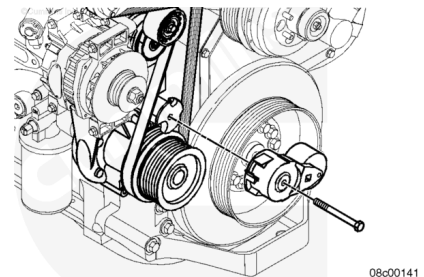
Check that the bottom tensioner arm stop is in contact with the bottom tensioner arm stop boss on the tensioner body. If these two are **not** contacting, the tensioner **must** be replaced.



Remove

Drives Alternator

Remove the mounting capscrew and the belt tensioner from the bracket.

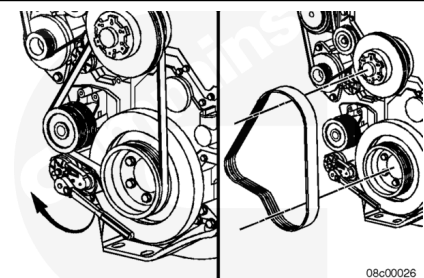


Automotive Applications Without EGR

Using a 1/2-inch square drive, lift belt tensioner to relieve tension on the belt.

Remove the belt.

Note : The belt tensioner is spring-loaded and **must** be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner.



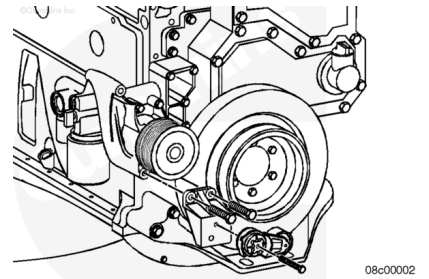
Low-Mount:

Note : Different engine configurations utilize different belt tensioner bracket and belt tensioner

arrangements.

Remove the belt tensioner mounting capscrew and belt tensioner from the bracket.

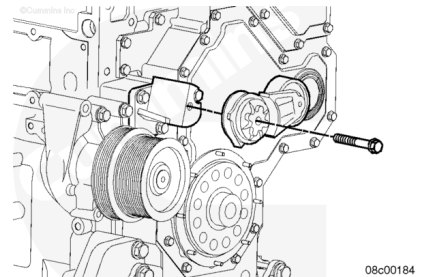
Remove the belt tensioner bracket from the water pump.



Mid-Mount:

Remove the belt tensioner mounting capscrew and the belt tensioner from the bracket.

Note : The mid-mount aluminum belt tensioner/idler bracket is part of the water pump removal and installation due to common capscrew usage. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/10/10-008-062.html>).

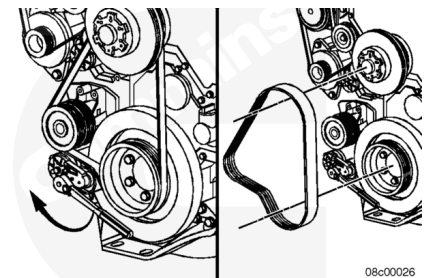


Industrial and Power Generation Application Without EGR

Using a 1/2-inch square drive, lift belt tensioner to relieve tension on the belt.

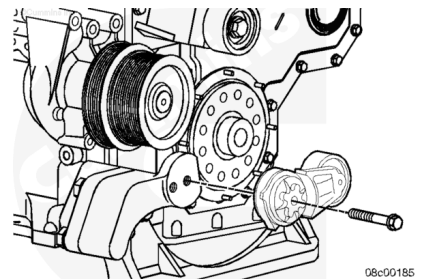
Remove the belt.

Note : The belt tensioner is spring-loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner.

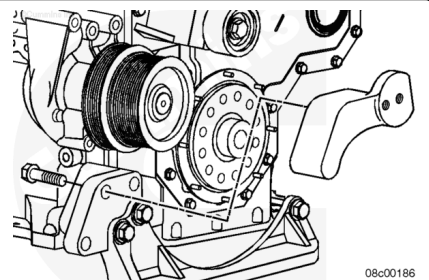


Low-Mount (Small Version)

Remove the belt tensioner mounting capscrew and belt tensioner from the bracket.

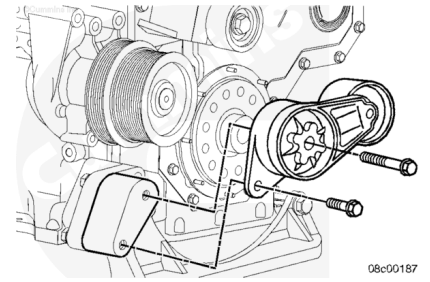


Remove the belt tensioner bracket from the front engine support.

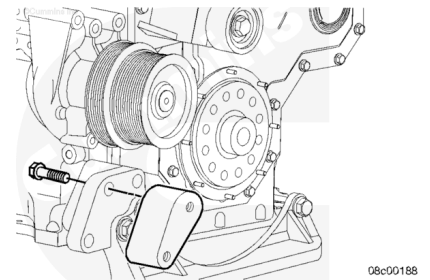


Low-Mount (Large Version)

Remove the belt tensioner mounting capscrew and belt tensioner from the bracket.



Remove the belt tensioner bracket from the front engine support.



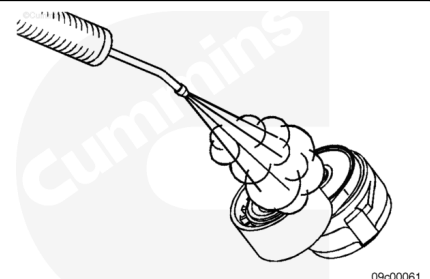
Inspect for Reuse

Drives Alternator

⚠ WARNING ⚠

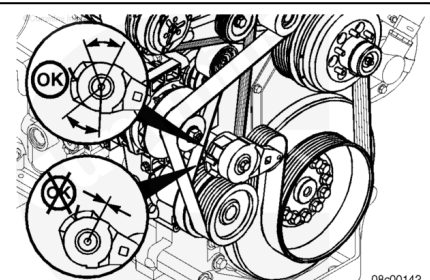
The belt tensioner is spring loaded. Do not attempt to disassemble the tensioner. Severe bodily injury can result.

Check the tensioner for dirt buildup. If the tensioner has dirt buildup, remove and clean with safety solvent.



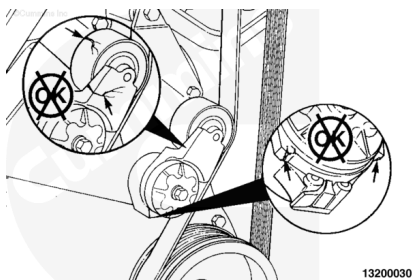
Pivot the tensioner through the full range of motion. Tensioner **must** move with resistance while pivoting the tensioner down. If **not**, replace the tensioner.

Check and make sure the correct belt part number is being used.

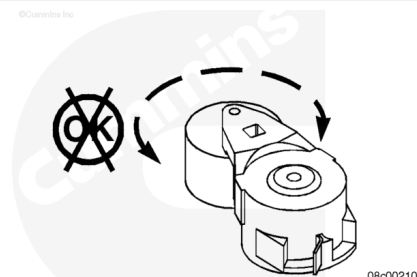


Check the belt tensioner pulley and body for cracks or damage. If any are found, replace the tensioner.

Inspect the tensioner arm stop bosses, for cracks or damage. If cracks or damage is found, replace the tensioner.

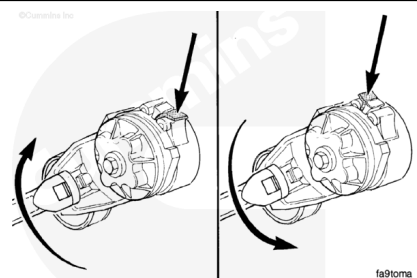


Spin the tensioner pulley. If pulley does **not** spin freely, replace the tensioner.

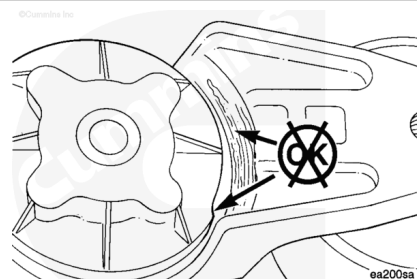


Without EGR

Check the tensioner for dirt buildup. If this condition exists, the tensioner **must** be removed and steam-cleaned.



Inspect the tensioner for evidence of the pivoting tensioner arm contacting the stationary circular base. If there is evidence of these two areas contacting, the pivot tube bushing has failed and the tensioner **must** be replaced.



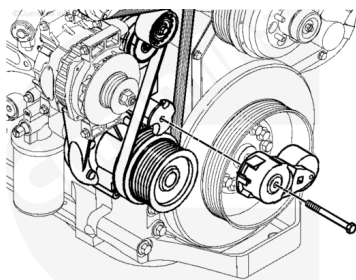
Install

Drives Alternator

Install the belt tensioner and mounting capscrew.

Tighten the capscrew.

Torque Value: 50 n•m [37 ft•lb]



08c00141

Automotive Applications Without EGR

Low-Mount

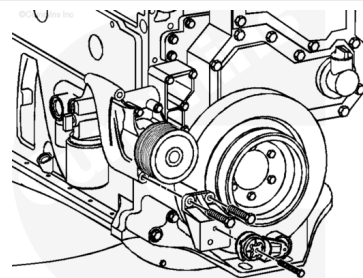
Install the belt tensioner bracket to the water pump.

Tighten capscrew.

Torque Value: 34 n•m [25 ft-lb]

Install the belt tensioner and capscrew.

Torque Value: 34 n•m [25 ft-lb]

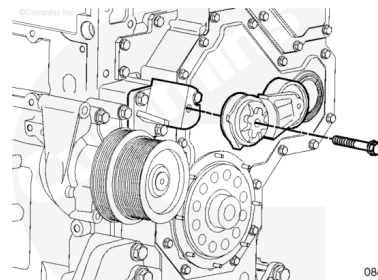


08c00002

Mid-Mount:

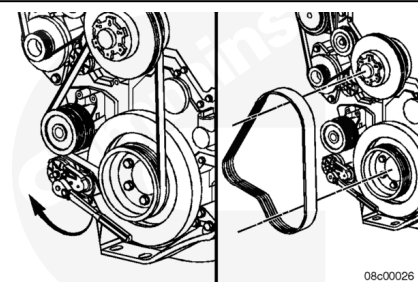
Install the belt tensioner mounting capscrew and the belt tensioner from the bracket.

Note : The mid-mount aluminum belt tensioner/idler bracket is part of the water pump removal and installation due to common capscrew usage. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/10/10-008-062.html>).



08c00184

Using a 1/2-inch square drive, lift and hold the tensioner. Install the water pump/fan drive belt and release the tensioner.



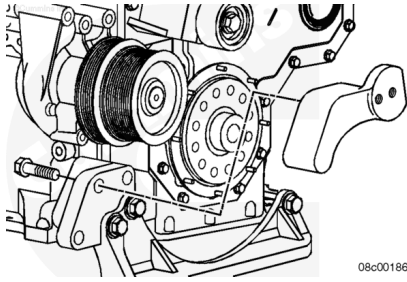
08c00026

Industrial and Power Generation Application Without EGR

Low-Mount (Small Tensioner)

Install the belt tensioner bracket to the front engine support with three mounting capscrews.

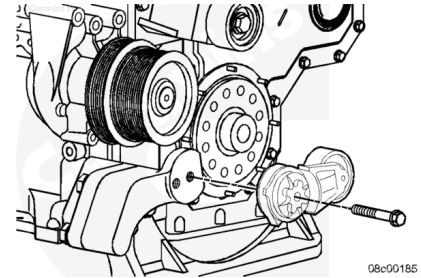
Torque Value: 56 n•m [41 ft-lb]



08c00186

Install the belt tensioner.

Torque Value: 56 n•m [41 ft-lb]

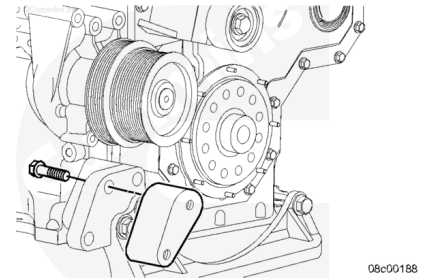


08c00185

Low-Mount (Large Tensioner)

Install the belt tensioner bracket to the front engine support with three mounting capscrews.

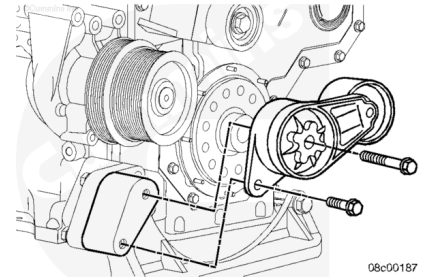
Torque Value: 56 n•m [41 ft-lb]



08c00188

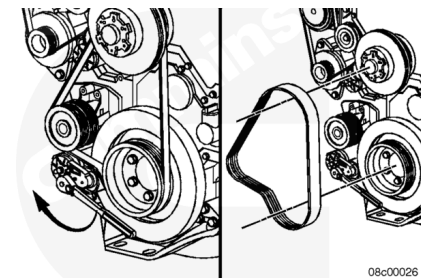
Install the belt tensioner.

Torque Value: 56 n•m [41 ft-lb]



08c00187

Using a 1/2-inch square drive, lift and hold the tensioner. Install the water pump/fan drive belt and release the tensioner.



08c00026

Finishing Steps

Drives Alternator

CAUTION

The belt tensioner is spring loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner or personal injury.

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Install the drive belt. Refer to Procedure 008-062
(</qs3/pubsys2/xml/en/procedures/10/10-008-062.html>).